Initial Project Idea (IPI)

MSc Applied Research Project (7WCM2038)
University of Hertfordshire

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Project Title

Evaluating Agentic Approaches for Legal Question Answering

Project Description

Legal AI tools increasingly rely on Retrieval-Augmented Generation, but real-world experience building AI Reads Law suggests it often performs poorly on legal questions and legal language differs from common speech, and similarity search returns noisy results. Newer agentic approaches (planning, reflection, tool use, multi-agent) may handle complex legal questions better, but there is no clear evidence of which works best. This project builds an evals to benchmark retrieval-augmented generation, agentic approaches, and hybrids on UK legal tasks. Technologies: Python, Jupyter, PostgreSQL. Research question: which combination of retrieval and reasoning techniques produce the most accurate answers to UK legal questions, and at what cost?

Project Tasks and Deliverables

Development work:

1. Curate a test set of UK legal questions covering simple lookup, multi-step reasoning, and ambiguous interpretation;
2. Build an evaluation pipeline that runs each technique against the test set and scores accuracy against ground truth, uses LLM as a judge, latency, and cost;
3. Implement and benchmark a representative range of techniques, including a keyword baseline, standard retrieval-augmented generation, and agentic variants. Deliverables: a reusable evaluation pipeline; a ranked comparison of techniques with cost and accuracy trade-offs; a literature review of at least thirty sources; a final report with practical recommendations for legal AI builders. Specific techniques benchmarked will be finalised during the project to reflect the fast-moving field.

Expected Supervisor Information

This idea was developed independently, based on the student's experience building AI Reads Law. It overlaps loosely with Canvas proposals #19 (LLMs in mobile apps), #26 (knowledge graphs), and #48 (LLM-powered adaptive systems), but does not match any listed topic directly. A supervisor with expertise in information retrieval, natural language processing, or applied large language model research would be ideal.